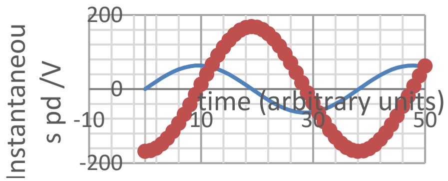
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*	(i)	Pd of L and C are opposite signs (antiphase) (or in phasor diag) (1) Pd of L 90° before $\frac{R}{I}$ and C 90° after $\frac{R}{I}$ (or in phasor diag) (1) Pd of supply is the resultant of all 3 (or in phasor diag) (1)	3			3		
		(ii)	Resonance: R pd is power supply pd (1) Pds of L and C are equal and opposite (1)	2			2		
	(b)	(i)	$\frac{119.1}{45} = 2.65$		1		1	1	
		(ii)	$\frac{45}{28} = 1.61$ [A]		1		1	1	
		(iii)	$X_C = \frac{V}{I}$ used [74.1 Ω] or $X_C = \frac{1}{\omega C}$ (1) Answer = 26.2 [Hz] (1)		2		2	2	
		(iv)	$X_L = \omega L$ used with 74.1 (1) **ecf** Answer = 0.45 [H] (1)	1	1		2	2	
		(v)	Peak V of 168 V (1) Phase as shown (1) 		2		2	1	
		(vi)	Reactance of inductor increases or reactance of capacitor decreases or $X_L > X_C$ (1) Impedance increases because reactances don't cancel (1)		2		2		

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
	(c)		Realising frequency variation of capacitor reactance e.g. decreases with frequency or small at high frequency or large at low frequency or explaining $\frac{1}{\omega C}$ (1) Voltage divider argument or equivalent employed leading to low pd at low f and vice versa (1) Calculating reactance at 20 kHz [43.01 Ω] (1) Realising $\sqrt{V_\mathrm{R}^2 + V_\mathrm{C}^2}$ = 12 and not 6 V + 6 V = 12 V (1) Correct conclusion Sam is wrong on pds (or should be 8.5 V) but OK on frequency relationship (1)			5	5	3	
			Question 6 total	**6**	**9**	**5**	**20**	**10**	**0**